

Academic Year: 2025 – 2026

IFET COLLEGE OF ENGINEERING

(An Autonomous Institution)

Department of Artificial Intelligence & Data Science

TinyML-Based Real-Time Motorcycle Crash Detection and Emergency Notification System Using ESP32 and Cellular Communication

SUPERVISOR

Mrs.P.DIVYA,
AP/AI&DS.

Presented By

M.D.CHARAN (421122106007)
A.DINESHKUMAR (421122106013)

INTRODUCTION

- Domain encompasses **Embedded Systems**, and **Intelligent Transportation Systems (ITS)**
- Road safety is a major concern due to increasing motorcycle accidents
- Delay in emergency response leads to higher fatality rate
- Riders may be unable to send alert after crash
- Existing safety solutions are costly and not portable
- Embedded systems enable real-time monitoring in vehicles
- Cellular communication provides long-range alert capability
- TinyML enables fast crash detection with low power consumption
- Proposed system aims to provide automatic accident detection and notification

ABSTRACT

- Manual accident reporting in 108 system causes delay in rescue
- Threshold-only detection leads to false crash alerts
- Proposed system uses threshold + TinyML classification
- MPU6050 used for motion sensing
- TinyML classifies normal, bump, brake, crash
- GSM + GPS send automatic alert with location link
- Control room dashboard shows real-time accident status
- Low-cost system for faster emergency response
- The project supports SDG 3 (**Good Health and Well-Being**) by helping faster emergency response

OBJECTIVE

- To develop an intelligent embedded system for real-time motorcycle accident detection
- To eliminate manual dependency in emergency reporting
- To reduce delay in emergency response after accidents
- To implement TinyML-based on-device decision making
- To improve accuracy using multi-sensor data analysis
- To minimize false alerts using a delay-based validation mechanism
- To automatically send alerts and calls to contacts and 108
- To enhance rider safety and survival chances

LITERATURE REVIEW

Title	Author	Content	Methodology
IoT-Based Accident Monitoring System (2024)	Sharma & Malhotra	Developed an IoT-based system that monitors accident conditions and sends alerts through connected platforms in real time.	IoT-based monitoring
Hybrid ML-Based Accident Detection System (2024)	Patel & Joshi	Proposed a hybrid system combining rule-based logic and machine learning to improve accident detection accuracy.	Hybrid rule-based + ML
Intelligent Accident Detection System (2024)	Kim & Choi	Designed an intelligent system using smart sensors to identify accident scenarios more accurately.	Intelligent detection
Threshold-Based Accident Detection (2025)	Hudati	Implemented a threshold-based approach using sensor values to detect accidents, though prone to false alerts.	Threshold-based method
IoT-Based Accident Alert System (2025)	Yadavrao et al.	Created an IoT-based system that sends accident alerts using embedded devices and communication modules.	IoT alert system

EXITING SYSTEM

➤ 108 Ambulance System

- Depends on manual phone call
- Fails if rider is unconscious or injured

➤ Smartphone-based applications

- Require user interaction
- Depend on internet connectivity
- May fail if phone is damaged during accident

➤ Delay in emergency response

EXITING SYSTEM

- Threshold-based detection systems
 - Use fixed sensor limits
 - Generate false alerts (braking, potholes)
- Lack of intelligent decision-making
- No multi-sensor data analysis
- No false alert validation mechanism
- Overall systems are unreliable in real-world conditions

LIMITATIONS OF EXISTING SYSTEM

- Dependence on manual reporting (108 system)
- Not effective if rider is unconscious or severely injured
- Delay in emergency response
- Threshold-based detection → high false alerts
- Smartphone-based systems
- Depend on internet connectivity
- Lack of intelligent decision-making
- No real-time automated accident detection

PROPOSED SYSTEM

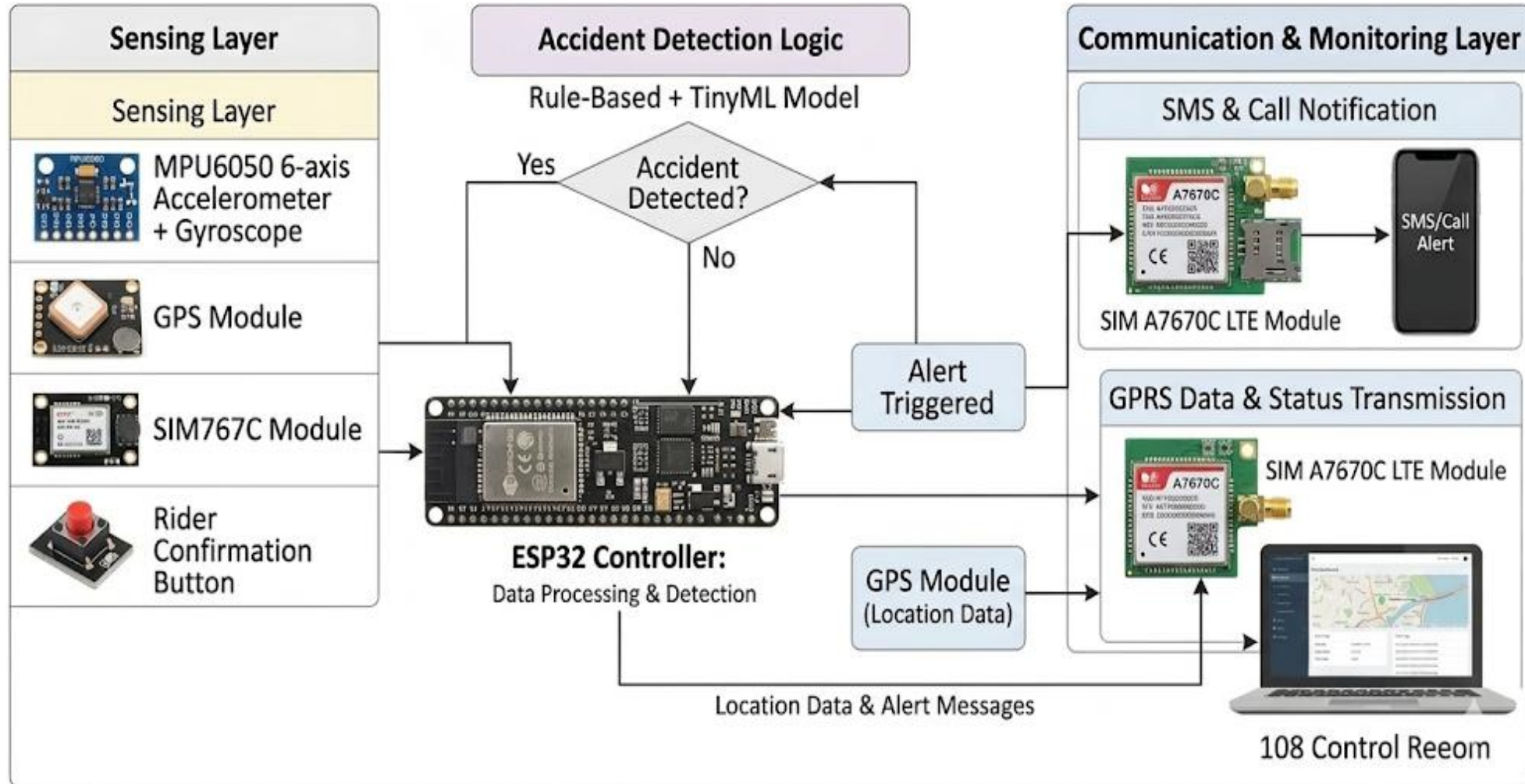
- Intelligent embedded system for real-time accident detection
- Uses motion sensors
 - Detects impact, tilt, abnormal movement
- TinyML model on ESP32
 - Local processing (no internet required)
 - Improves accuracy
- Uses multi-sensor data fusion

Distinguishes normal vs accident conditions

PROPOSED SYSTEM

- Delay-based alert mechanism
 - User can cancel false alerts
- If no response →
 - Automatic SMS + Call
 - Sent to contacts & 108
- Ensures fast emergency response
- Reduces false alerts
- Provides reliable and autonomous safety system

SYSTEM ARCHITECTURE



MODULE DESCRIPTION

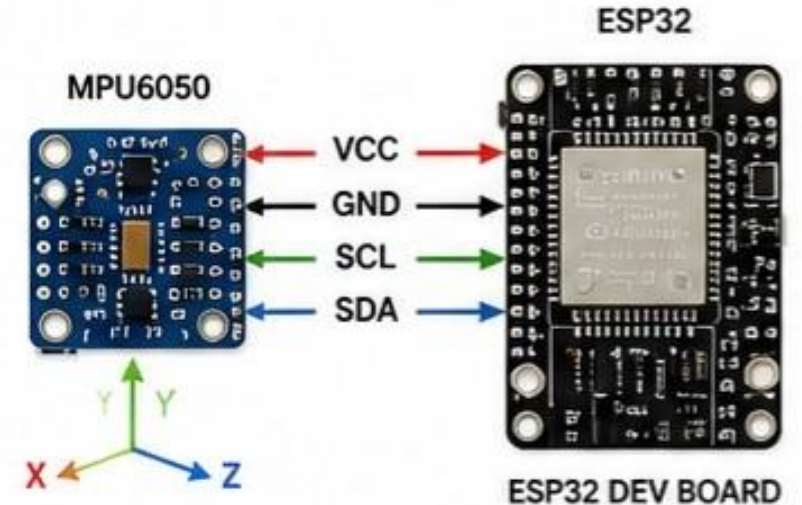
1. Sensor Data Acquisition Module:

Module uses the **MPU6050 sensor** to continuously acquire real-time motion parameters such as acceleration (X, Y, Z axes), angular velocity, and tilt orientation. The collected data reflects the dynamic behavior of the motorcycle and is essential for identifying abnormal conditions such as sudden impact, skid, or fall.

1 SENSOR DATA ACQUISITION MODULE

Function:

Collects real-time motion, tilt, and impact data from the vehicle.



Components Used:

- MPU6050 Accelerometer + Gyroscope
- ESP32 Dev Board



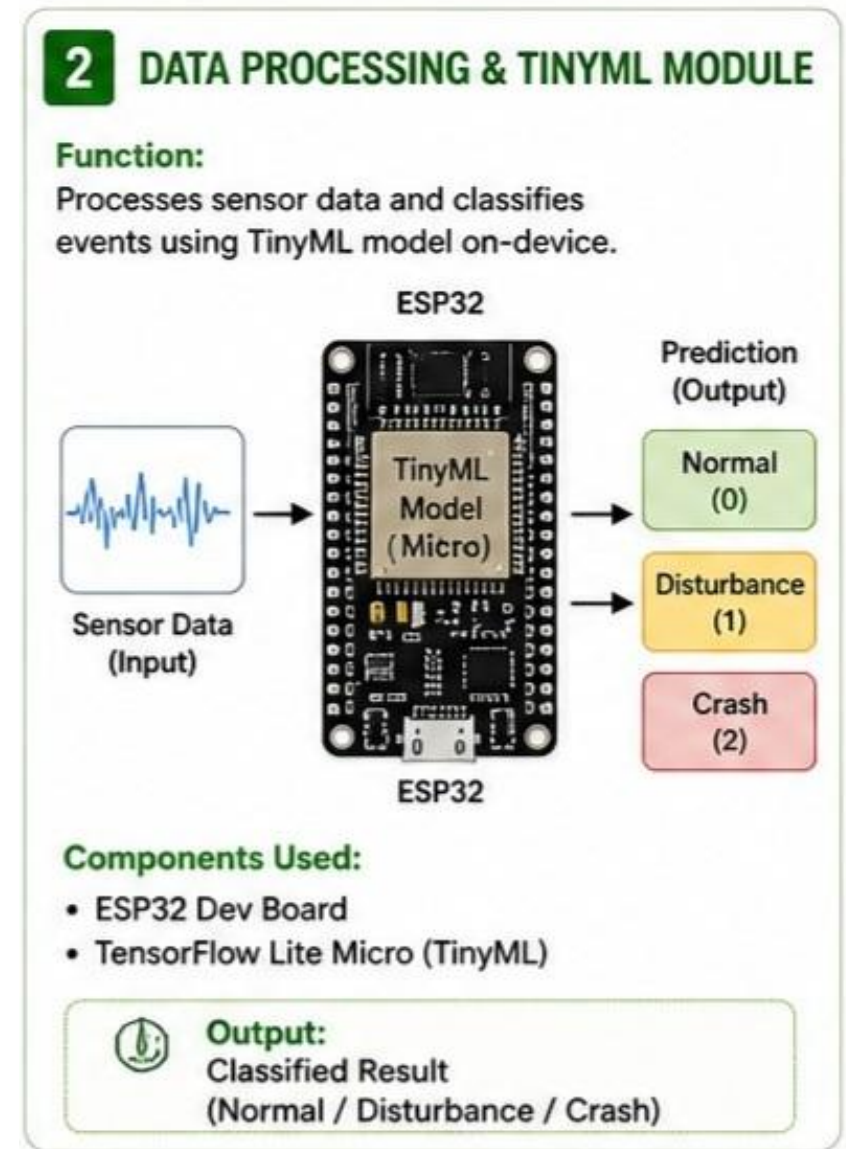
Output:

Raw sensor data
(Acceleration, Gyroscope, Tilt)

MODULE DESCRIPTION

2. Data Processing & TinyML Module:

The acquired sensor data is processed locally on the ESP32 microcontroller using a TinyML model deployed through TensorFlow Lite Micro. The model analyzes patterns in motion data to classify events into categories such as normal riding, disturbance, or crash conditions.



MODULE DESCRIPTION

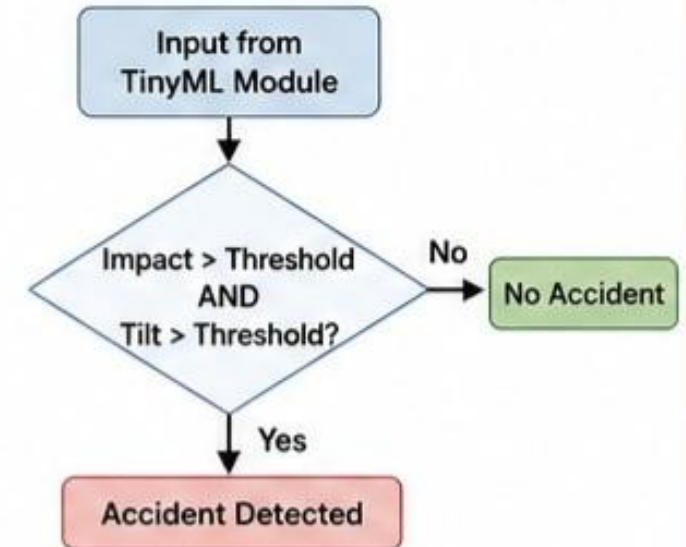
3. Accident Detection Module:

The acquired sensor data is processed locally on the ESP32 microcontroller using a TinyML model deployed through TensorFlow Lite Micro. The model analyzes patterns in motion data to classify events into categories such as normal riding, disturbance, or crash conditions.

3 ACCIDENT DETECTION MODULE

Function:

Detects real accident conditions based on ML output and rule-based threshold logic.



Components Used:

- ESP32 (Logic + ML Output)

Output:

Accident Detected Signal

MODULE DESCRIPTION

4. Alert Validation Module (Timer System):

Upon detecting a potential accident, a predefined delay timer is activated. During this interval, the rider is provided with an option to cancel the alert using a push button. If no action is taken within the time window, the system confirms the event as a true emergency.

4 ALERT VALIDATION MODULE (TIMER SYSTEM)

Function:

Provides time window to user to cancel false alerts before sending.



Components Used:

- Push Button
- ESP32 (Timer Control)

Output:

Confirmed Alert / Cancelled Alert

MODULE DESCRIPTION

5. Communication Module (GSM & 108 Integration):

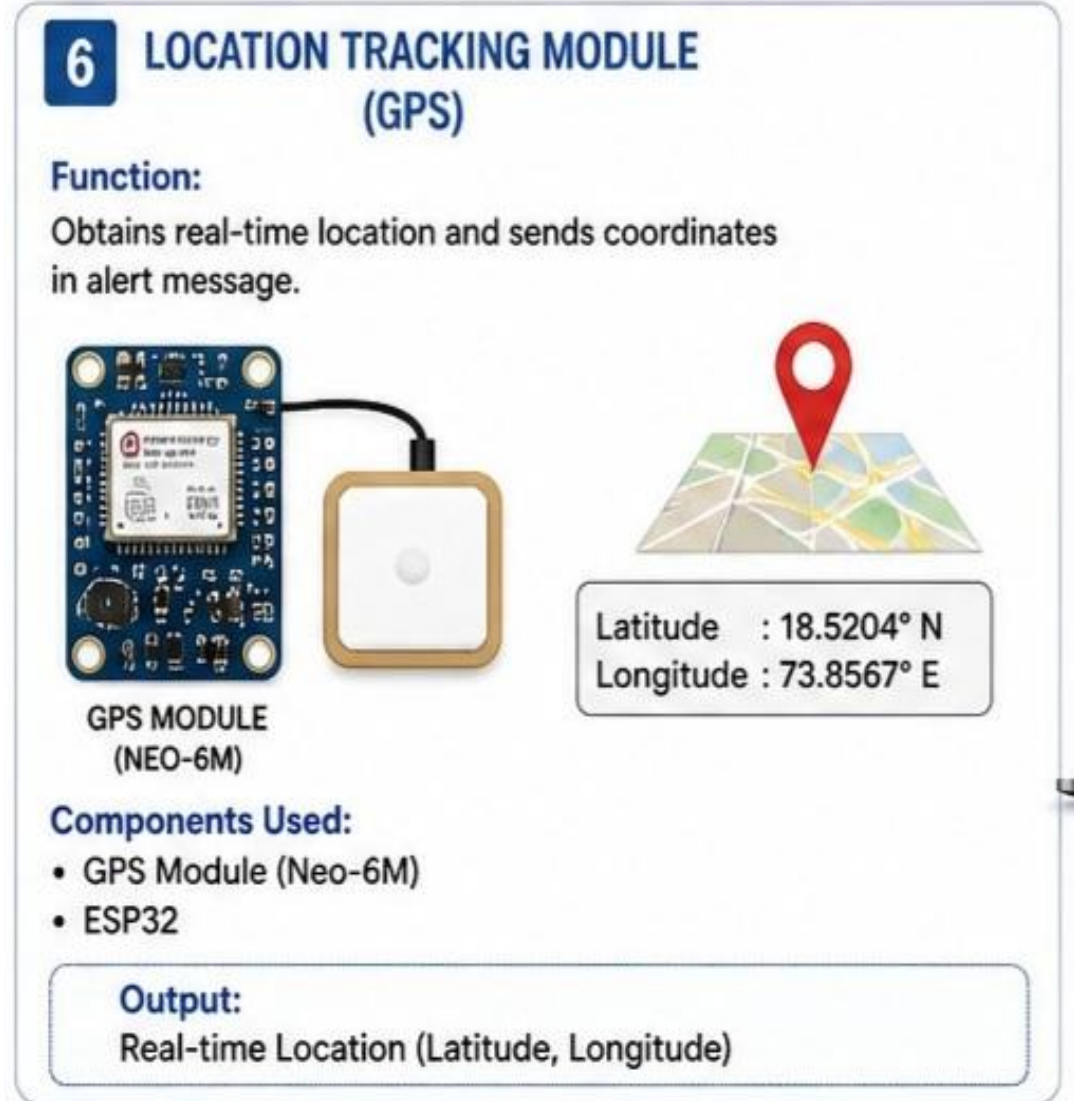
Once the alert is confirmed, the GSM module (SIM A7670C) is triggered to send SMS alerts and initiate calls to predefined contacts. The system can also dial emergency services such as **108**, ensuring immediate assistance. This module operates independently of internet connectivity, making it reliable in remote areas.



MODULE DESCRIPTION

6. Location Tracking Module (GPS):

The GPS module retrieves real-time latitude and longitude coordinates of the accident location. These coordinates are appended to the emergency message sent via GSM, allowing responders to quickly identify the exact location of the incident.



MODULE DESCRIPTION

7. Monitoring & Dashboard Module:

A web-based dashboard developed using Flask provides real-time monitoring of system status, alert logs, and communication outputs. It acts as a control interface for tracking system performance and verifying alert transmissions during testing and deployment.

7 **MONITORING & DASHBOARD MODULE**

Function:
Displays system status, alerts, and logs in real-time on a web dashboard.

Recent Alerts

ID	Status	Time	Location
12	Accident	10:30:45 AM	18.5204° N, 73.8567° E
11	Cancelled	10:15:20 AM	18.5187° N, 73.8541° E
10	Accident	09:45:10 AM	18.5190° N, 73.8600° E

Components Used:

- Laptop / PC
- Flask (Web Server)

Output:
Real-time Monitoring & Alerts on Dashboard

REQUIERMENTS

Software Tools:

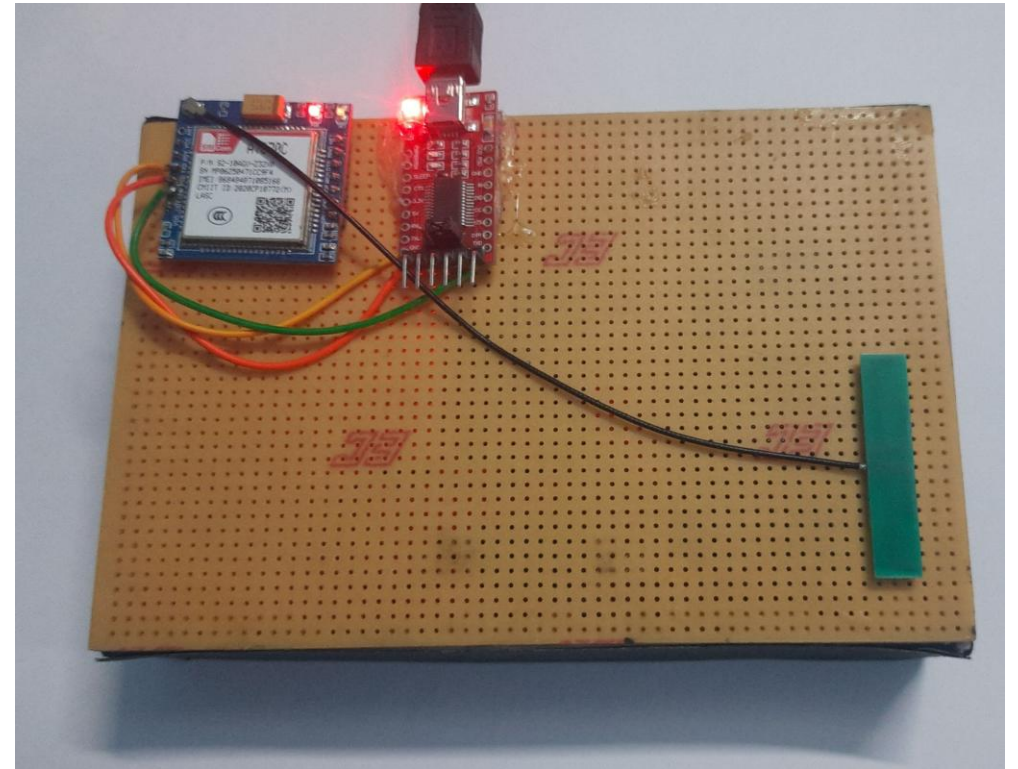
- Arduino IDE – for programming ESP32
- ESP32 Libraries – sensor and GSM communication handling
- TinyML – for crash classification using machine learning
- TensorFlow Lite Micro – for TinyML model execution
- Web Server Library (ESP32) – for dashboard monitoring interface
- Python / Flask – for control room dashboard
- HTML / CSS / JavaScript – for dashboard UI design

Hardware Components:

- ESP32 Microcontroller – main controller
- MPU6050 Sensor – motion and impact detection
- SIM A7670C GSM/LTE Module – SMS + network communication
- GPS Module – location tracking
- Buzzer – accident alert sound
- Push Button – cancel alert option
- Battery – power supply
- Laptop / PC – dashboard monitoring
- USB-TTL Converter – control room GSM connection

PROTOTYPE

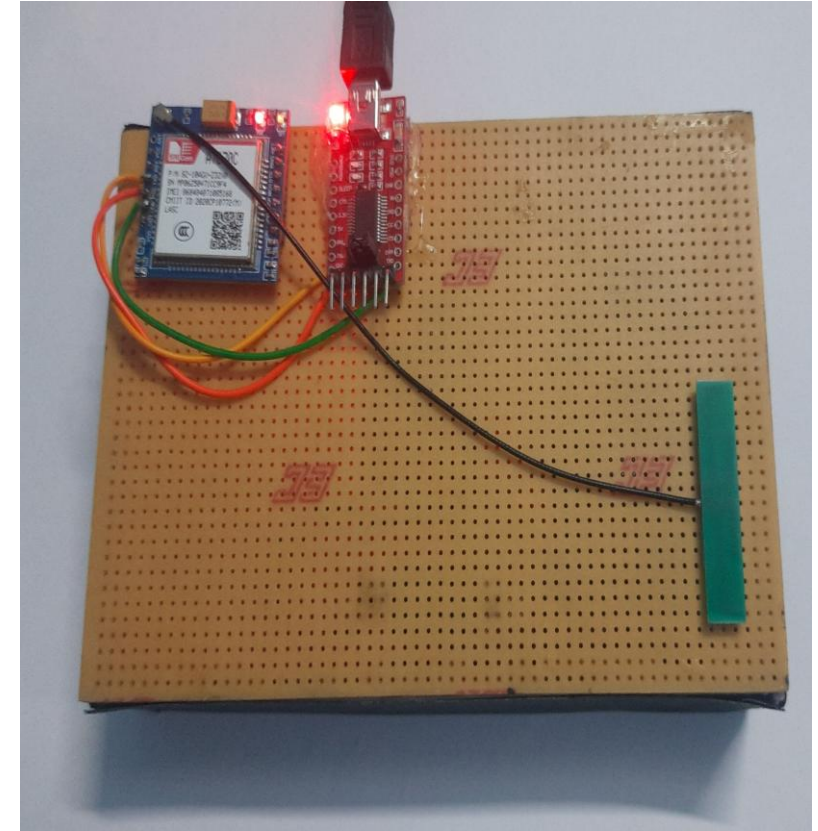
- Installed on the motorcycle prototype for real-time operation
- Collects motion data using MPU6050 (acceleration & tilt)
- Processes data using ESP32 with TinyML model
- Accurately detects accident conditions (impact + tilt)
- Activates timer for user validation (cancel option)
- Sends SMS and call alerts with location to contacts & 108



Vehicle Unit

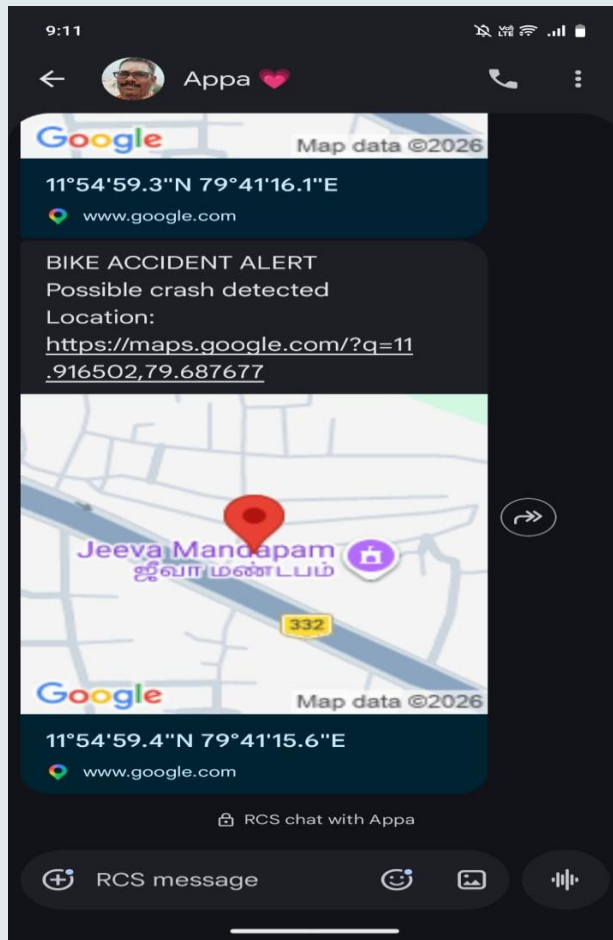
PROTOTYPE

- Acts as control and monitoring system
- Receives alerts via GSM communication
- Displays information on web-based dashboard
- Shows accident status, time, and location
- Used for real-time monitoring and system validation
- Stores and tracks alert logs for analysis



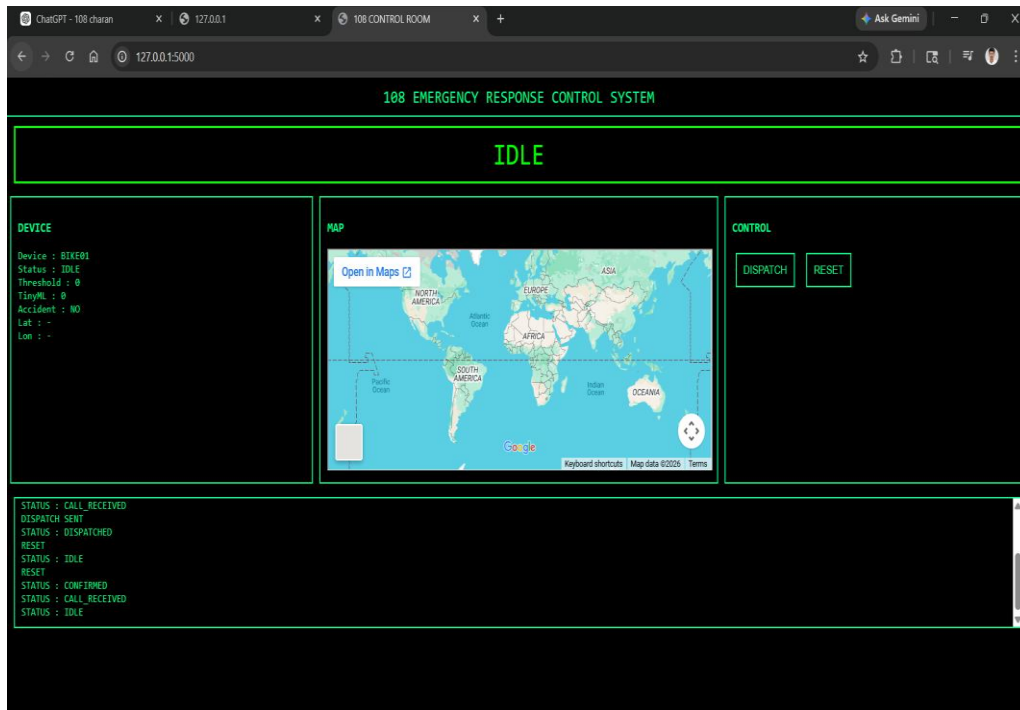
Receiver Unit

RESULTS & DISCUSSION



- System successfully detects accident conditions in real-time
- Automatically sends emergency SMS alert to predefined contact
- Message includes crash notification and exact GPS location
- Location is shared as a Google Maps link for easy navigation

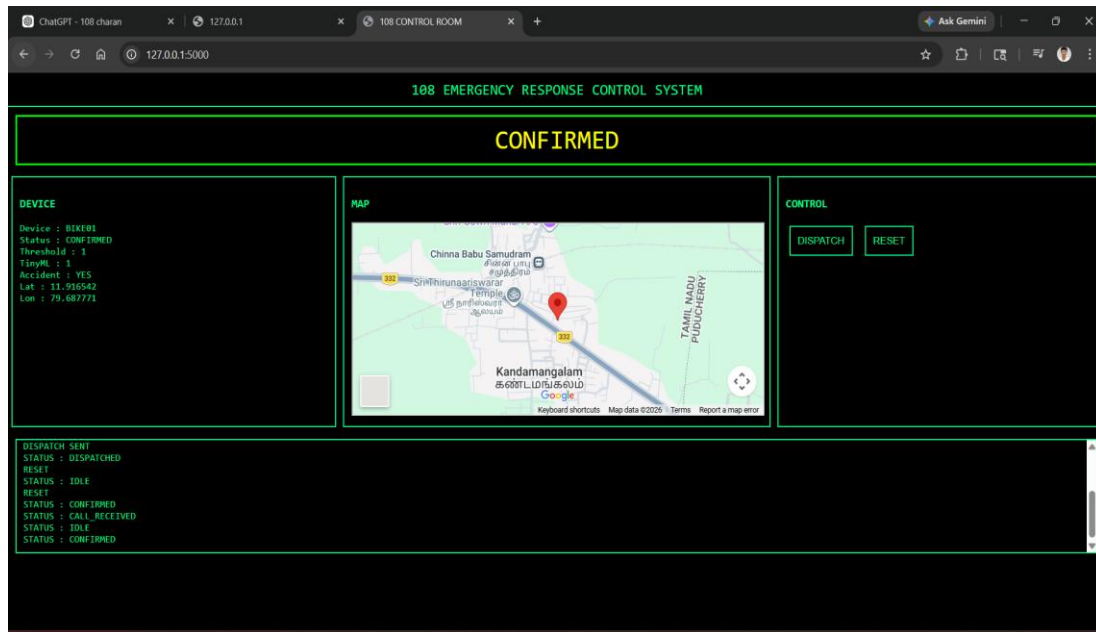
RESULTS & DISCUSSION



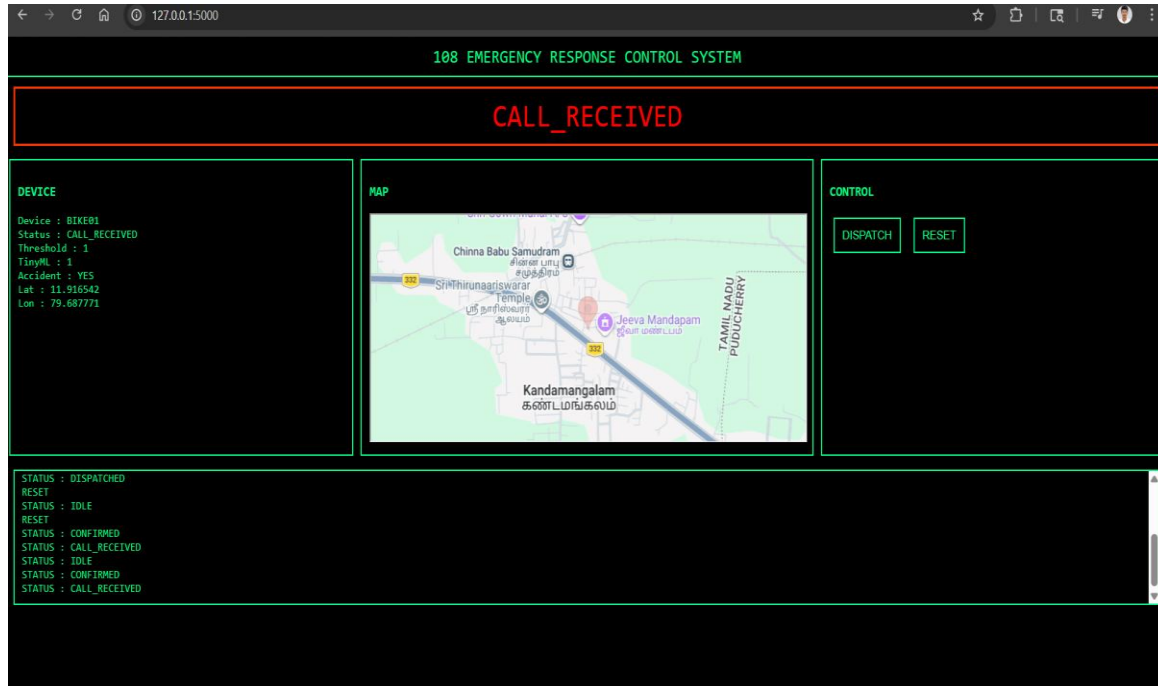
- System is in normal monitoring condition
- No accident detected (Accident = NO)
- Device is actively sending live status updates
- Map is ready but no incident location displayed
- Waiting for accident detection signal
- Dashboard shows stable system operation

RESULTS & DISCUSSION

- SMS is received from vehicle unit
- TinyML + logic confirms true accident condition
- System status changes to CONFIRMED
- Accurate location is displayed on map
- Ready for emergency dispatch action
- Reduces false alerts before response

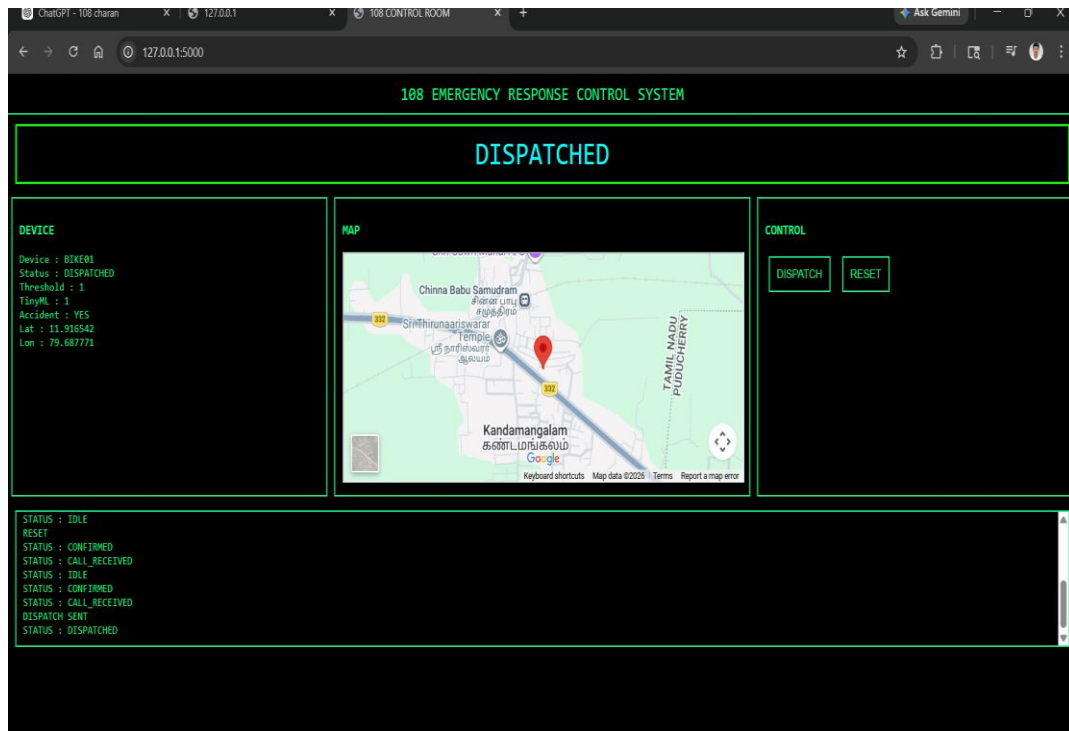


RESULTS & DISCUSSION



- Accident detected and alert is received
- System status changes to CALL_RECEIVED
- Location data (latitude & longitude) is updated
- Map displays accident location marker
- Indicates incoming emergency event
- Awaiting confirmation or further action

RESULTS & DISCUSSION



- Emergency response is successfully triggered
- System sends dispatch signal
- Status updated to DISPATCHED
- Location shared with responders
- Control unit initiates rescue process
- Confirms end-to-end system functionality

CONCLUSION

- Developed an intelligent embedded system for accident detection
- Eliminates manual dependency in emergency reporting
- Uses TinyML for real-time, on-device decision making
- Improves accuracy using multi-sensor data analysis
- Reduces false alerts with timer-based validation
- Automatically sends alerts + calls to contacts and 108
- Ensures faster emergency response
- Provides a cost-effective and scalable safety solution
- Enhances rider safety and survival chances

REFERENCES(CHICAGO)

- [1] Barhate, Minal Yadavrao, Soham Milind Lonkar, Aditya Namdev Mali, Arjun Vishwas Mane, Samidha Appasaheb Mandage, Shivam Dhanaji Mane, and Swara Mamidwar. "IoT-Enabled Vehicle Accident Detection System Using ESP32 with GPS Tracking and Dual-Channel Emergency Alerts via GSM-SMS and Telegram Bot." In *1st Engineering Data Analytics and Management Conference (EAMCON 2025)*, pp. 623-635. Atlantis Press, 2025..
- [2] Hudati, Tirthankar, and Jasonnita Das. "Accident Detection System Using GPS GSM Module and Arduino." *IJSAT-International Journal on Science and Technology* 16, no. 3 (2025).
- [3] Ravindran, Santhiya, and Gurukarthik Babu Balachandran. "Next generation road safety solutions: IoT-driven accident prevention for smart riders." *Journal of Industrial Information Integration* 44 (2025): 100804.
- [4] Sahraei, Mohammad Ali, and Said Ramadhan Mubarak Al Mamari. "A review of Internet of Things approaches for vehicle accident detection and emergency notification." *Sustainability* 17, no. 14 (2025): 6510